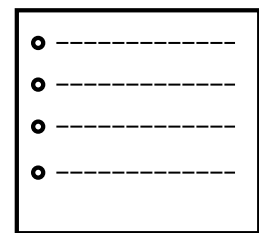


Clinical variables & epidemiology



patient histories, age, sex, FBCs and NLR

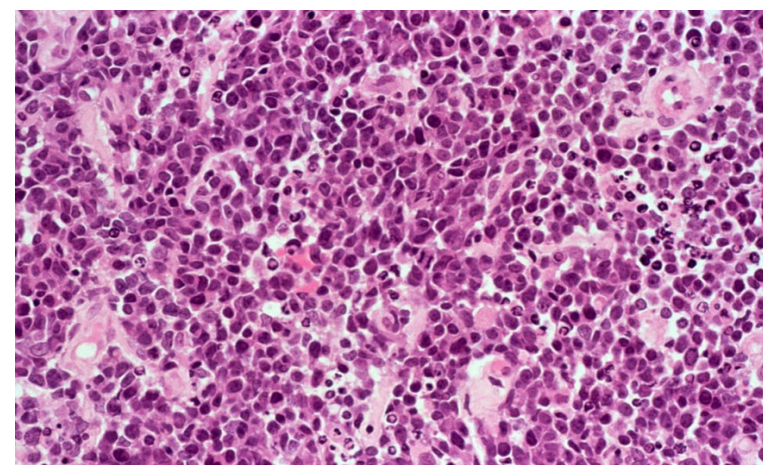
age, type,
38.30, 3,
18.27, 4,
43.64, 4,

...

...

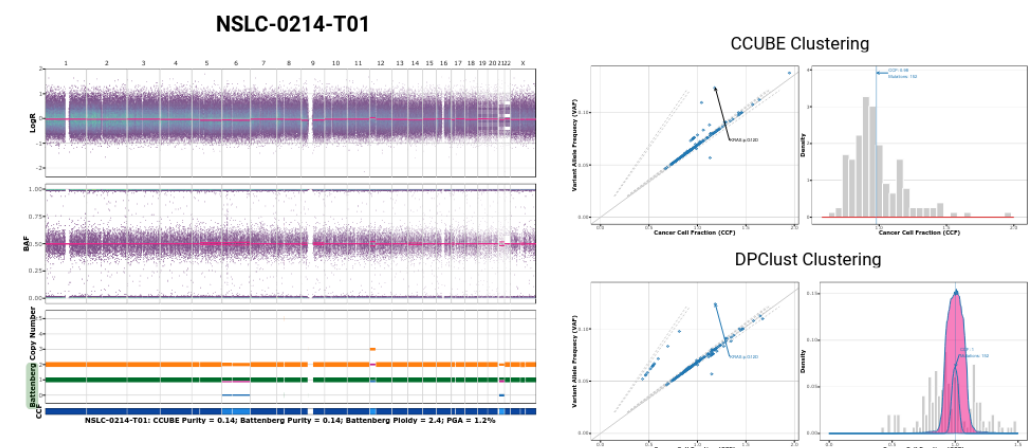
...

H&E slides and imaging



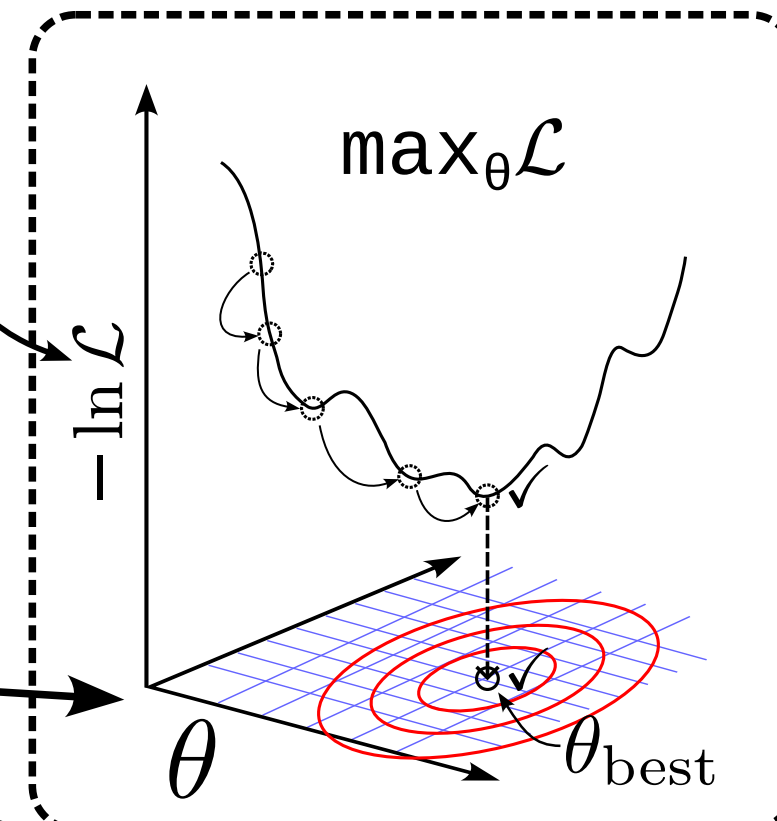
Whole genome sequencing

Supplementary Fig. 30

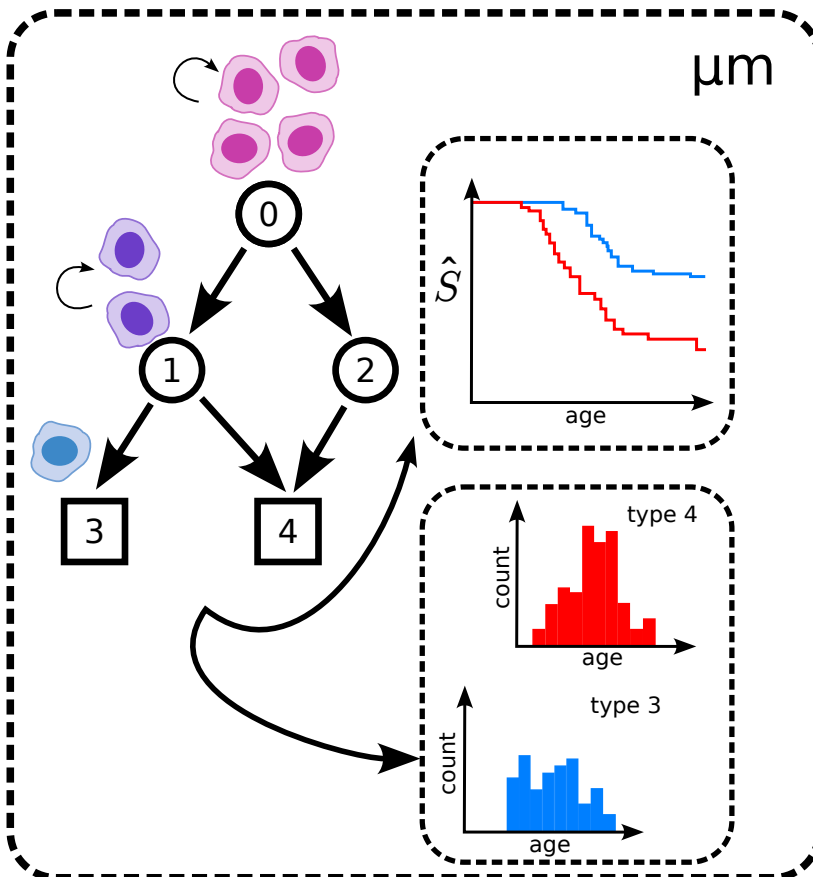


all data modalities contribute to cost function (log-likelihood)

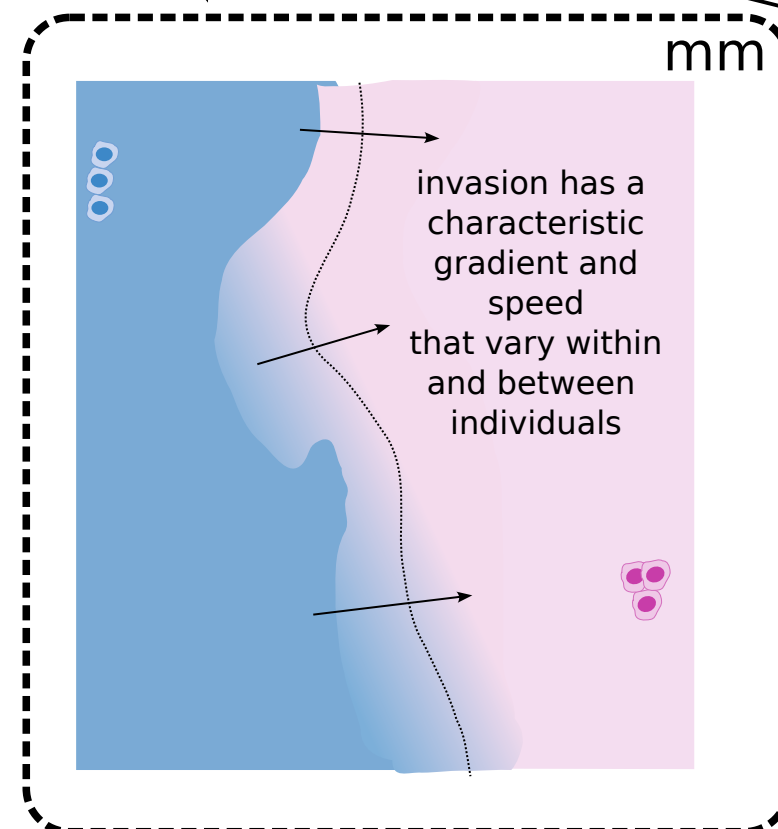
all scale models have a shared parameter space Θ



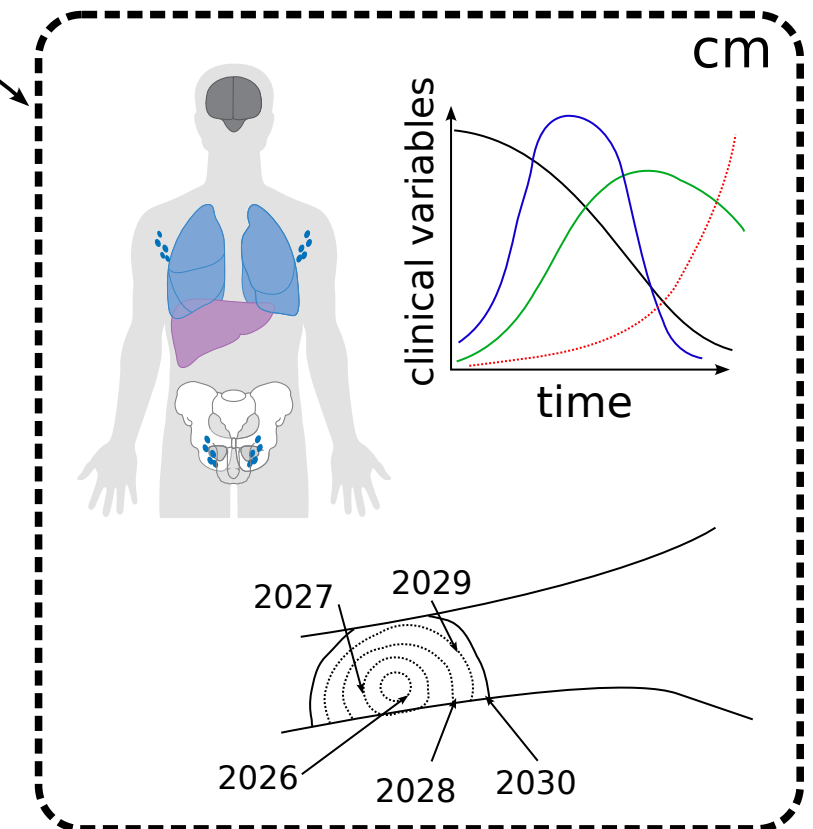
learning algorithm optimises parameters of all submodels simultaneously



niche scale



tissue scale



systemic scale